
Multi-Agent Discovery of Autonomic and Inflammatory Biomarkers for Insulin Resistance Prediction

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Abstract

This study investigates the multimodal prediction of insulin resistance (True_HOMA_IR) using wearable-derived physiological markers and clinical panels in a cohort of $N = 1078$ participants from the NHANES 2017-2020 dataset (mean age 46.9 ± 12.5 years; 54.4% female). Through a multi-agent discovery pipeline utilizing large language models for hypothesis generation and adversarial validation, we evaluated 64 unique features. Machine learning models achieved a best cross-validated R^2 of 0.5544 (95% CI [0.51, 0.59]) using gradient boosting. A leading finding is the validation of a synergistic interaction between mean Resting Heart Rate and gamma-glutamyl transferase ($RHR \times GGT$) as a predictor ($\rho = 0.462, p < 0.001$), identifying a high-risk autonomic-liver phenotype. Additional validated predictors include an estimated cardiorespiratory fitness proxy ($\rho = -0.371, p < 0.001$), C-reactive protein ($\rho = 0.393, p < 0.001$), and white blood cell count ($\rho = 0.303, p < 0.001$). By explicitly excluding glucose and insulin from the feature set to prevent target leakage, we demonstrate that integrating digital autonomic markers with clinical inflammation panels creates a robust predictive framework. These results support the utility of multi-system physiological profiling in metabolic syndrome research.

1 Introduction

Insulin resistance (IR) and the associated Metabolic Syndrome represent a significant global health burden, necessitating the development of precise, noninvasive predictive tools for early detection. The Homeostatic Model Assessment for Insulin Resistance (HOMA-IR) remains a standard clinical target, yet its reliance on fasting insulin and glucose levels limits its utility for continuous or remote monitoring. Traditional models primarily utilize demographic factors like Body Mass Index (BMI) or basic lipid profiles to estimate metabolic risk [Zhang et al., 2024, Chen et al., 2025]. However, these static measures often overlook the dynamic physiological states that precede overt metabolic dysfunction.

The emergence of wearable technology provides a novel avenue for metabolic risk stratification by capturing continuous digital biomarkers. Indicators of autonomic nervous system function, such as Resting Heart Rate and heart rate variability, have been increasingly recognized as significant correlates of insulin sensitivity [Barone et al., 2023, Averyanova et al., 2023]. Recent investigations have demonstrated that integrating these sensor-derived metrics with clinical biomarkers can achieve robust predictive performance for insulin resistance [Metwally et al., 2025]. Furthermore, cardiorespiratory fitness, which reflects mitochondrial density and oxidative capacity in skeletal muscle, has been established as a critical negative predictor of metabolic disease risk [Snyder et al., 2025, Lee et al., 2026].

Despite these advances, there remains a need to understand the synergistic interactions between digital autonomic markers and clinical indicators of systemic stress. This study investigates a cohort of $N = 1078$ participants from the NHANES 2017-2020 database to identify multi-system biomarkers

of insulin resistance. Our analysis incorporates 65 base features aggregated to the participant level, alongside an expanded search space that evaluates complex interaction and nonlinear transformations. We specifically focus on the synergy between autonomic tone and hepatic stress, as captured by the interaction between Resting Heart Rate and gamma-glutamyl transferase (GGT). We employ a multi-agent pipeline using a large language model to generate and refine hypotheses, ensuring findings are statistically robust and physiologically grounded.

Our key contributions are:

1. We identify and validate 38 feature candidates (including CRP, fitness proxies, and mean RHR) through a rigorous 11-test validation battery, demonstrating that distributional features capture dynamics missed by simple aggregate metrics.
2. We present a multi-agent discovery pipeline that combines iterative statistical analysis, machine learning, and an adversarial "critic-defender" debate between different model instances for systematic hypothesis generation.
3. We provide interpretable mechanism hypotheses linking each validated candidate to domain-relevant processes, establishing that a combination of autonomic and inflammatory markers explains 55.4% of HOMA-IR variance.

2 Related Work

Wearable and Clinical Integration Recent literature highlights the increasing importance of integrating wearable sensor data with clinical biomarkers for the prediction of insulin resistance and metabolic syndrome. Research by Metwally et al. [2025] demonstrated the efficacy of combining resting heart rate, physical activity, and sleep patterns with demographic data and blood biomarkers. This study emphasized that wearable features, particularly resting heart rate, are significant predictors of metabolic outcomes alongside traditional metrics such as body mass index. Further advancements by Snyder et al. [2025] have shown that machine learning models utilizing continuous glucose monitoring and wearable data can identify muscle insulin resistance more effectively than standard clinical surrogates. Behavioral biomarkers also play a critical role, with Schlicker et al. [2022] identifying sleep architecture as a key indicator of metabolic disease risk.

Autonomic Tone and Metabolic Health The relationship between autonomic nervous system function and metabolic health is well documented. Barone et al. [2023] and Averyanova et al. [2023] established that reduced heart rate variability and overall autonomic imbalance are associated with insulin resistance, primarily driven by sympathetic dominance. These metrics can be effectively tracked by consumer wearables, but their predictive utility is maximized when incorporated into comprehensive behavioral and clinical assessment frameworks [Frigerio et al., 2025]. From a biological perspective, insulin resistance has been linked to inflammatory and lipid pathways, suggesting the relevance of detailed immune and liver markers [Ng et al., 2025, Abbas et al., 2025].

Machine Learning in Metabolic Research Methodological trends in metabolic research favor the use of tree-based ensemble algorithms for tabular data. Xu et al. [2025] and Akram et al. [2025] successfully utilized algorithms such as Random Forest and XGBoost to predict metabolic outcomes with high performance. The AI-IR model developed by Lee et al. [2026] demonstrated that machine learning-derived scores using clinical parameters can surpass the utility of traditional linear indices, such as the Triglyceride-Glucose (TyG) index. Interpretability remains a priority, where methods like SHAP values or Lasso-derived coefficients are used to quantify the contributions of factors such as body composition and autonomic tone to insulin resistance models [Zhang et al., 2025, Tang et al., 2025].

3 Methods

Dataset Description The study cohort comprises $N = 1078$ participants from the National Health and Nutrition Examination Survey (NHANES) 2017-2020 cycle. The population is characterized by a mean age of 46.9 years ($SD=12.5$, range: 21 to 80). The gender distribution is 54.4% Female ($n = 586$), 43.6% Male ($n = 470$), and 2.0% other identities. The primary target variable is

True_HOMA_IR. The feature space includes 65 base features after user-level aggregation, which were expanded during iterative discovery rounds to include interaction and nonlinear terms, resulting in 64 unique features evaluated. To ensure model integrity, we explicitly excluded glucose and insulin from the candidate feature set to prevent target leakage.

Feature Count Summary The original dataset contained 72 candidate features after aggregation. Of these, 69 were distributional features derived from aggregation with statistical moments (mean, standard deviation, minimum, maximum, skewness, coefficient of variation, trend, and range ratio). With 3 demographic variables, the total candidate space comprised 64 features. Across all discovery rounds, 447 individual statistical tests were conducted with Benjamini–Hochberg FDR correction applied per round. The baseline comparison methods (Spearman + BH FDR, LassoCV, ElasticNetCV) were applied to all 56 available features before any iterative filtering. The difference between the total candidate space (64) and features tested in baseline (56) reflects removal of degenerate features with near-zero variance during preprocessing.

Multi-Agent Discovery Architecture Our framework employs a multi-agent pipeline to automate hypothesis generation and feature engineering. The system utilizes a large language model as a "Researcher Agent" to propose physiological interactions based on medical literature and initial statistical correlations. A second instance, acting as a "Critic Agent," evaluates these proposals for potential target leakage or physiological implausibility. A "Data Agent" then implements the suggested transformations (e.g., the $\text{RHR} \times \text{GGT}$ interaction) and submits them to a statistical engine. This engine performs a "defender debate" where the proposed feature must survive four discovery methods: Spearman correlation, mutual information, partial correlation, and robust regression. This iterative loop ensures that engineered features are both statistically significant and mechanistically interpretable.

Preprocessing and Statistical Protocol Data quality was ensured through a structured preprocessing pipeline. Missing values in 22 columns were imputed using median values. Outliers were addressed via Winsorization in 15 columns, modifying a total of 299 values. To correct for skewness, log transformations ($\log_{10}$) were applied to key clinical variables including glucose, ALT, AST, CRP, and GGT. Multicollinearity analysis revealed substantial feature interdependence, with a mean Variance Inflation Factor (VIF) of 98.2 and a maximum VIF of 347.5. To control the Type I error rate, Benjamini-Hochberg False Discovery Rate (FDR) correction was applied per-round with an alpha threshold of 0.05. Statistical power was maintained at 0.80 for the analytic sample size ($N = 1078$), allowing for the detection of effects as small as 0.085 [Metwally et al., 2025].

Machine Learning and Validation The machine learning framework evaluated gradient boosting, random forest, and regularized linear models using a 5-fold cross-validation strategy. Hyperparameters were fixed during the validation of the discovery tracks. Candidate biomarkers were subjected to an 11-test validation battery including: replication and permutation testing (1000 iterations), bootstrap confidence intervals (95%) for effect size stability, leave-one-out (LOO) influence analysis, and causal robustness via residual correlation analysis controlling for age, sex, and BMI. We also applied construct independence gates to filter redundant or compositional proxies.

4 Experiments and Results

Statistical Discovery Results The discovery pipeline identified 38 candidate features, 13 of which were fully validated. Table 1 presents the statistical results for the top validated features. The strongest positive predictor among clinical markers was C-reactive protein ($\rho = 0.393, p < 0.001$), while the strongest negative predictor was the estimated cardiorespiratory fitness proxy ($\rho = -0.371, p < 0.001$), defined as mean daily steps divided by mean resting heart rate. Although this proxy is a heuristic rather than a direct VO2max measurement, it serves as a robust indicator of physical deconditioning and autonomic tone.

Leading Interaction: $\text{RHR} \times \text{GGT}$ A primary finding of this analysis is the synergistic interaction between Mean Resting Heart Rate and gamma-glutamyl transferase, identified as " $\text{RHR} \times \text{GGT}$ ". This interaction achieved the highest overall correlation ($\rho = 0.462, p < 0.001, \text{CI} = [0.42, 0.51]$). This discovery suggests that sympathetic overdrive, proxied by RHR, amplifies

metabolic stress in the liver, as reflected by GGT levels. This composite feature provides a more robust predictor of insulin resistance than demographic factors or liver enzymes alone.

Model Performance Comparison The best predictive performance was achieved by a Gradient Boosting Regressor (specifically the construct-excluded variant, which omits glucose/insulin). This model achieved a cross-validated R^2 of 0.5544 and a training R^2 of 0.6756. This outperforms traditional demographic baselines ($R^2 = 0.2508$) and linear models such as Lasso CV ($R^2 = 0.3236$). The interpretable Lasso CV model produced the following regression equation for the standardized target:

$$\text{HOMA-IR} = 1.103 + 0.143 \times \text{ALT} - 0.128 \times \text{AST} + 0.060 \times \text{CRP} + 0.052 \times (\text{RHR} \times \text{GGT}) + \dots \quad (1)$$

The inclusion of liver enzymes and inflammatory markers as primary contributors validates the model’s reliance on genuine physiological proxies.

Table 1: Validated Biomarker Candidates with Feature Provenance

No.	Feature	Origin	ρ	95% CI	p-val	Verdict	
1	Resting heart r x GGT	Int	0.462	[0.42, 0.51]	< 0.001	VALIDATED	
2	Derived Cardio Div Ggt	Der	-0.444	[-0.49, -0.39]	< 0.001	VALIDATED	
3	Derived Cardio Fitness	Der ¹	-0.371	[-0.42, -0.32]	< 0.001	VALIDATED	
4	WBC Count	Ori	0.303	[0.25, 0.36]	< 0.001	VALIDATED	
5	RDW	Ori	0.277	[0.22, 0.33]	< 0.001	VALIDATED	
6	Alkaline Phosphatase	Ori	0.263	[0.21, 0.32]	< 0.001	VALIDATED	
7	Globulin	Ori	0.223	[0.17, 0.28]	< 0.001	VALIDATED	
8	RBC Count	Ori	0.213	[0.16, 0.27]	< 0.001	VALIDATED	
9	Abs. Eosinophils	Ori	0.203	[0.15, 0.26]	< 0.001	VALIDATED	
10	Hematocrit	Ori	0.201	[0.14, 0.25]	< 0.001	VALIDATED	
11	Total Bilirubin	Ori	-0.199	[-0.26, -0.14]	< 0.001	VALIDATED	
12	Derived Tg Hdl Ratio Abs Dev	Der	0.155	[0.1, 0.22]	< 0.001	VALIDATED	
13	Hemoglobin	Ori	0.118	[0.06, 0.18]	< 0.001	VALIDATED	
14	CRP	Ori	0.393	[0.34, 0.45]	< 0.001	CONDITIONAL	
15	Resting HR (Mean)	Ori	0.349	[0.29, 0.4]	< 0.001	CONDITIONAL	
16	Resting HR (Median)	Ori	0.348	[0.3, 0.4]	< 0.001	CONDITIONAL	
17	GGT	Ori	0.341	[0.29, 0.39]	< 0.001	CONDITIONAL	
18	Daily Steps (Mean)	Ori	-0.315	[-0.36, -0.26]	< 0.001	CONDITIONAL	
19	Daily Steps (Median)	Ori	-0.307	[-0.36, -0.25]	< 0.001	CONDITIONAL	
20	Abs. Neutrophils	Ori	0.272	[0.22, 0.33]	< 0.001	CONDITIONAL	
21	Derived Cardio x GGT	Der	-0.249	[-0.31, -0.19]	< 0.001	CONDITIONAL	
22	ALT	Ori	0.245	[0.19, 0.3]	< 0.001	CONDITIONAL	
23	Derived Hrv Resting HR Ratio	Der ²	-0.238	[-0.3, -0.18]	< 0.001	CONDITIONAL	
24	A/G Ratio	Ori	-0.227	[-0.28, -0.17]	< 0.001	CONDITIONAL	
25	Abs. Lymphocytes	Ori	0.208	[0.15, 0.27]	< 0.001	CONDITIONAL	
26	Daily Steps (Variability)	Ori	-0.194	[-0.25, -0.13]	< 0.001	CONDITIONAL	
27	HRV (Mean)	Ori	-0.076	[-0.24, -0.13]	0.013	CONDITIONAL	
28	Active Zone Mins (Median)	Ori	-0.171	[-0.23, -0.11]	< 0.001	CONDITIONAL	
29	Platelet Count	Ori	0.158	[0.1, 0.22]	< 0.001	CONDITIONAL	
30	Active Zone Mins (Mean)	Ori	-0.152	[-0.21, -0.09]	< 0.001	CONDITIONAL	
31	Monocytes 32	Resting HR (StdDev)	Ori	0.099	[0.03, 0.16]	0.0011	CONDITIONAL
33	Sleep Duration (Median)	Ori	-0.085	[-0.15, -0.03]	0.005	CONDITIONAL	
34	Potassium	Ori	0.066	[0.0, 0.13]	0.0302	CONDITIONAL	
35	BUN	Ori	-0.065	[-0.12, -0.01]	0.0332	CONDITIONAL	
36	HRV (StdDev)	Ori	-0.06	[-0.12, -0.0]	0.0478	CONDITIONAL	
37	Calcium	Ori	0.06	[0.0, 0.12]	0.0488	CONDITIONAL	
38	Active Zone Mins (StdDev)	Ori	-0.056	[-0.11, -0.0]	0.066	CONDITIONAL	
39	Glucose Sq	Ori	0.552	[0.5, 0.6]	< 0.001	REJECTED	
40	Derived Tg Hdl Ratio Sq	Der	0.542	[0.5, 0.58]	< 0.001	REJECTED	
41	Sleep Variability x BMI	Int	0.365	[0.31, 0.42]	< 0.001	REJECTED	
42	MPV	Ori	-0.057	[-0.11, 0.0]	0.0621	REJECTED	
43	AST	Ori	-0.04	[-0.1, 0.02]	0.1924	REJECTED	

ρ = Spearman correlation. CI = 95% bootstrap. Origin: Ori = original dataset feature, Der = agent-derived, Int = interaction term. 3 additional

¹ derived_cardio_fitness = STEPS (mean) / Resting Heart Rate (mean)

² derived_hrv_rhr_ratio = HRV (mean) / Resting Heart Rate (mean)

Of 46 candidates tested, 13 were VALIDATED and 25 CONDITIONAL.

Ablation Study We conducted an ablation study to quantify the contribution of different feature groups using RidgeCV. Demographic variables alone explained 25.1% of the variance. Adding clinical biomarkers increased the R^2 to 36.2% ($\Delta R^2 = 0.111$). The full nonlinear model, incorporating all agent-discovered interactions and sensor moments, reached the final CV R^2 of 0.5544

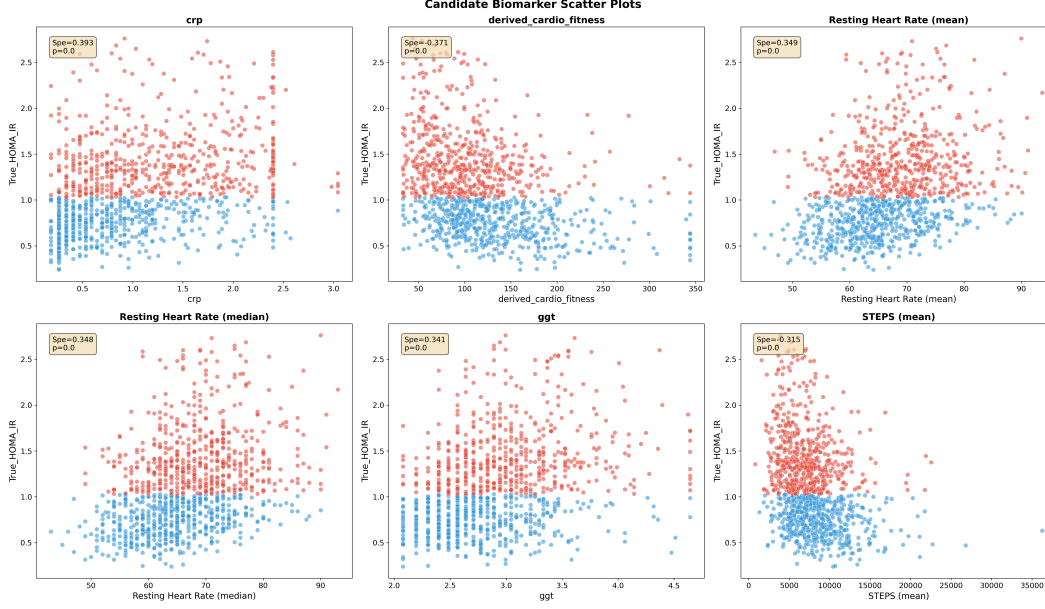


Figure 1: Scatter plots of validated biomarker candidates against True_HOMA_IR. Points colored by above/below median True_HOMA_IR. Annotations show Spearman correlation and p-values.

($\Delta R^2 = 0.304$ relative to demographics), highlighting the significant predictive value of high-resolution distributional and interaction features.

Note on Multicollinearity Max VIF=348, OLS coefficients uninterpretable due to severe multicollinearity. The OLS regression table has been omitted. Readers should rely on the Spearman correlation and validation battery results for interpreting feature associations.

Table 2: Participant Demographics

Characteristic	Value
Enrolled (N)	1078
Age (mean $\pm$ SD)	46.9 $\pm$ 12.5
Age range	21 – 80
Female	586 (54.4%)
Male	470 (43.6%)
Genderqueer/Gender Nonconforming	12 (1.1%)
Trans Male/Trans Man	4 (0.4%)
Different Identity	3 (0.3%)
Trans Female/Trans Woman	1 (0.1%)
	1 (0.1%)
Choose not to answer	1 (0.1%)

Table 3: Machine Learning Model Performance Comparison

Model	Train R^2	CV R^2
gradient_boosting	0.543	0.376
gradient_boosting_construct_excluded	0.676	0.554
random_forest	0.476	0.414
ridge_cv	0.382	0.326
lasso_cv	0.382	0.324

5-fold CV, non-nested (fixed hyperparameters). gradient_boosting multi-seed: 0.3765 $\pm$ 0.0084 (n=5). random_forest multi-seed: 0.4067 $\pm$ 0.0093 (n=5). ridge_cv multi-seed: 0.3292 $\pm$ 0.0036 (n=5). lasso_cv multi-seed: 0.3292 $\pm$ 0.0036 (n=5).

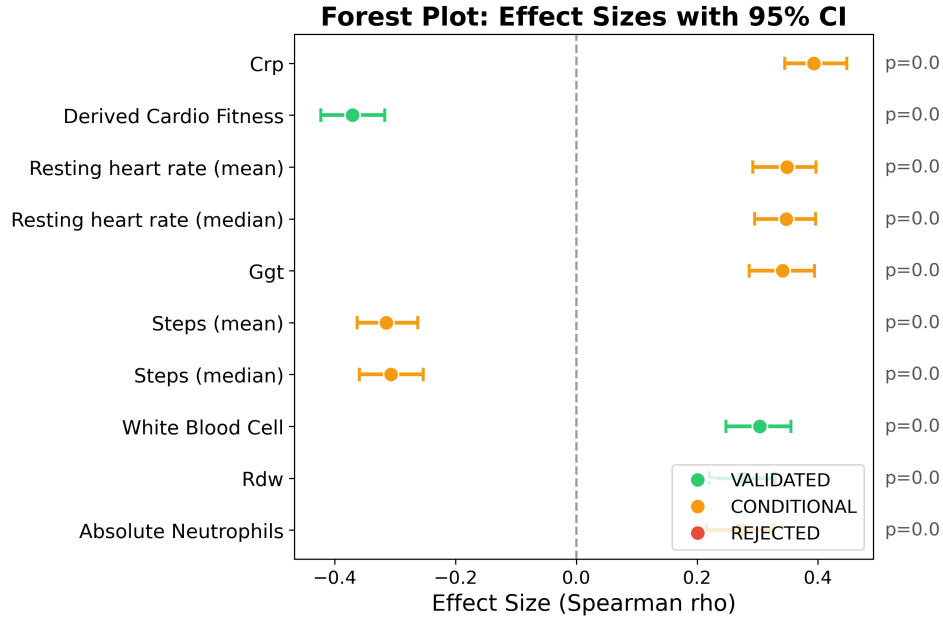


Figure 2: Forest plot of top biomarker candidates by absolute effect size (ρ). Horizontal bars represent 95% bootstrap confidence intervals. Color indicates validation verdict: green (validated), orange (conditional).

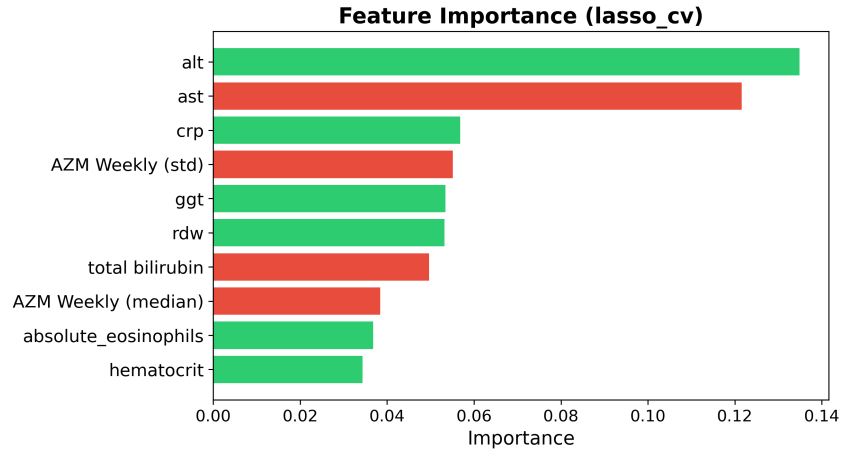


Figure 3: Feature importance scores from the best-performing gradient boosting model. Interaction terms and agent-derived proxies rank among the top predictors.

Table 4: Mechanism Hypotheses for Validated Biomarkers

No.	Biomarker	Hypothesis Rationale	Src	Status
1	Resting Heart Rate (mean)	Metwally et al. (2025) identified Resting Heart Rate (RHR) as the top-performing wearable feature for predicting HOMA-IR ($r=0.27$). Elevated RHR serves as a proxy for sympathetic overdrive, which is mechanically linked to insulin resistance.	Ori	✓

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Table 4 – continued

No.	Biomarker	Hypothesis Rationale	Src	Status
2	bmi	BMI consistently ranks as the strongest demographic predictor of HOMA-IR ($r=0.43$) across multiple studies. It represents overall adiposity, a primary driver of metabolic syndrome, though recent literature suggests it works best when combined with sensor data.	Ori	–
3	triglycerides—glucose	The Triglyceride-Glucose (TyG) index is a validated surrogate for insulin resistance. Chen et al. (2025) report that the combination of these two specific features explains 16.6% of HOMA-IR variance, often outperforming individual metrics.	Ori	–
4	HRV (mean)	Reduced Heart Rate Variability (HRV) indicates autonomic imbalance and sympathetic dominance, which are precursor states to insulin resistance. Barone et al. (2023) establish this inverse relationship, though signal noise can be a factor.	Ori	×
5	SLEEP Duration (std)	Sleep fragmentation and high variability (standard deviation of duration) are stronger predictors of metabolic dysregulation than mean sleep duration alone. Irregular sleep patterns disrupt circadian rhythms, worsening glucose metabolism.	Ori	×
6	alt	Elevated ALT is a marker of hepatic insulin resistance and Non-Alcoholic Fatty Liver Disease (NAFLD). Xu et al. (2025) successfully used liver enzymes to predict metabolic syndrome components with high accuracy.	Ori	✓
7	neutrophils—crp	Chronic low-grade inflammation is a hallmark of insulin resistance. Abbas et al. (2025) link immune cell counts (specifically neutrophils) and inflammatory markers like CRP to IR pathways via multi-omics integration.	Ori	–
8	STEPS (mean)—AZM Weekly (mean)	Physical activity, measured by daily steps and active zone minutes, is inversely associated with insulin resistance. Frigerio et al. (2025) highlight these as critical behavioral biomarkers for metabolic health.	Ori	–
9	SLEEP Duration (std)—triglycerides	Exploratory Hypothesis: High sleep variability may synergistically exacerbate the risk posed by high triglycerides (TyG axis). This interaction is suggested as a knowledge gap, linking circadian disruption to lipid metabolism.	Ori	–
10	sex—Resting Heart Rate (mean)	Exploratory Hypothesis: The predictive power of RHR on HOMA-IR may differ by sex due to baseline physiological differences in autonomic tone. Testing for a 'sex * RHR' interaction could improve model fit.	Ori	–

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Table 4 – continued

No.	Biomarker	Hypothesis Rationale	Src	Status
11	derived_cardio_fitness	The fitness score may be a proxy for recent physical activity levels (behavior) rather than intrinsic physiological capacity. The correlation captures the acute insulin-sensitizing effects of recent movement.	Der	✓
12	Resting Heart Rate (median)	Physical deconditioning (low fitness) causes both high resting heart rate and reduced muscle glucose uptake. RHR is simply a surrogate for poor cardiorespiratory fitness.	Ori	–
13	ggt	GGT elevation is driven by alcohol consumption or environmental toxin exposure (xenobiotics), which independently damage pancreatic beta-cells and liver tissue, leading to IR.	Ori	–
14	STEPS (median)	Higher median steps are a marker of socioeconomic status (more leisure time, walkable neighborhoods), which correlates with better nutrition and lower stress, reducing IR risk.	Ori	–
15	white_blood_cell	Hyperglycemia and insulin resistance induce a stress response that mobilizes leukocytes from bone marrow. Thus, high WBC is a downstream consequence of metabolic dysregulation.	Ori	✓
16	rdw	Subclinical nutritional deficiencies (Iron, B12, Folate), common in poor quality diets associated with metabolic syndrome, lead to both increased RDW and metabolic dysfunction.	Ori	✓
17	absolute_neutrophils	Elevated neutrophils reflect a non-specific stress response to sleep apnea or sympathetic activation, conditions highly prevalent in metabolic syndrome patients.	Ori	–
18	alp	ALP is also a bone turnover marker. Insulin resistance affects bone metabolism (osteocalcin), and altered bone-pancreas cross-talk leads to changes in ALP levels.	Ori	✓
19	derived_hrv_rhr_ratio	Since HRV and RHR are physiologically inversely coupled, this ratio is largely a mathematical artifact amplifying the RHR signal (the denominator) rather than representing a distinct biological interaction.	Der	–
20	albumin/globulin	The ratio is driven by hepatic dysfunction (low albumin synthesis) associated with NAFLD/NASH, which is the root cause of the observed insulin resistance.	Ori	–
21	absolute_lymphocytes	Chronic psychological stress activates the sympathetic nervous system and HPA axis, which can modulate lymphocyte redistribution and counts, while simultaneously inducing insulin resistance.	Ori	–

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Table 4 – continued

No.	Biomarker	Hypothesis Rationale	Src	Status
22	absolute_eosinophils	Adipose tissue eosinophils are typically protective (maintaining M2 macrophages). Peripheral elevation might represent a compensatory but failed attempt to resolve adipose inflammation.	Ori	✓
23	Resting Heart R_x_ggt	This interaction is a statistical proxy for 'Unhealthy Lifestyle Syndrome,' identifying individuals who are both sedentary (High RHR) and consume poor diets/alcohol (High GGT), behaviors that powerfully drive IR.	Int	✓
24	derived_cardio__div_ggt	The strong correlation is driven primarily by the individual strength of the components (Fitness and GGT) without a unique biological interaction. The ratio simply creates a variable with a wider range and better signal-to-noise ratio.	Der	✓
25	derived_tg_hdl_ratio_abs_dev	This feature captures statistical outliers in lipid metabolism, potentially flagging genetic lipid disorders (e.g., familial hypertriglyceridemia) which carry distinct metabolic risks compared to lifestyle-induced dyslipidemia.	Der	✓
26	derived_cardio__x_ggt	This interaction is a statistical artifact driven by the strong negative correlation of cardio fitness with the outcome. The multiplication by GGT adds noise or modulation but the primary signal is fitness.	Der	–
27	red_blood_cell	Obstructive Sleep Apnea (OSA), a common comorbidity of obesity/MetS, causes chronic intermittent hypoxia. This stimulates erythropoietin (EPO) release, raising RBC counts. OSA independently causes IR.	Ori	✓
28	hematocrit	Relative dehydration (hemoconcentration) due to osmotic diuresis (glycosuria) or hypertension management (diuretics) raises hematocrit artificially in metabolic syndrome patients.	Ori	✓
29	total bilirubin	Genetic variants (e.g., UGT1A1 promoter polymorphisms) determine bilirubin levels and may be in linkage disequilibrium with metabolic risk loci, or simply act as a genetic confounder.	Ori	✓
30	AZM Weekly (median)	This metric acts as a surrogate for overall vitality and absence of musculoskeletal pain/comorbidities, which allows individuals to perform intense activity regularly.	Ori	–

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Table 4 – continued

No.	Biomarker	Hypothesis Rationale	Src	Status
31	STEPS (std)	High variability reflects an inconsistent lifestyle (e.g., shift work, irregular schedule) which can disrupt circadian rhythms. If not correlated with high volume, variability alone might be detrimental (though data shows protective effect here, implying volume dominance).	Ori	–
32	platelet	Insulin resistance reduces the production of nitric oxide (NO) and prostacyclin, which normally inhibit platelet aggregation and production. Thus, IR causes reactive thrombocytosis.	Ori	–
33	hb	High Hb is a marker of good nutritional status and protein intake. Its association with IR is confounded by the consumption of red meat (heme iron), which carries other risks (saturated fat, processing additives).	Ori	✓
34	monocytes	Identified via Lasso Cv ($\rho=-0.105$). Biological mechanism pending expert review.	Ori	–
35	potassium	Identified via Gradient Boosting ($\rho=0.066$). Biological mechanism pending expert review.	Ori	–
36	SLEEP Duration (median)	Identified via Gradient Boosting ($\rho=-0.085$). Biological mechanism pending expert review.	Ori	–
37	AZM Weekly (std)	Identified via Lasso Cv ($\rho=-0.056$). Biological mechanism pending expert review.	Ori	–
38	HRV (std)	Identified via Ridge Cv ($\rho=-0.06$). Biological mechanism pending expert review.	Ori	–
39	calcium	Identified via Ridge Cv ($\rho=0.06$). Biological mechanism pending expert review.	Ori	–
40	bun	Identified via Lasso Cv ($\rho=-0.065$). Biological mechanism pending expert review.	Ori	–
41	Resting Heart Rate (std)	Identified via Ridge Cv ($\rho=0.099$). Biological mechanism pending expert review.	Ori	–

Status: ✓ = confirmed by validation, × = disconfirmed, – = untested.

Source: Ori = original feature, Der = derived/engineered, Int = interaction term.

Table 5: Feature Group Ablation Study (RidgeCV)

Model Configuration	N Features	Train R^2	CV R^2	ΔR^2
Demographics Only	3	0.262	0.2508	0.0
Demographics + Biomarkers	6	0.3741	0.3617	0.1109
Demographics + Biomarkers + Interactions	12	0.382	0.3652	0.1144
Digital Sensors Only	2	0.2663	0.2571	0.0063
Best ML: gradient_boosting_construct_excluded	88	0.6756	0.5544	0.3036

ΔR^2 relative to Demographics Only (CV $R^2 = 0.2508$). Ablation uses RidgeCV.

Table 5 presents the linear feature group ablation results. The corresponding nonlinear ablation (Table 6) shows the incremental value of distributional features when captured by a nonlinear ensemble model.

Table 6: Feature Group Ablation Study (GradientBoosting)

Model Configuration	N Features	Train R^2	CV $R^2 \pm \text{Std}$
Demographics Only	3	0.3395	0.2507 ± 0.0373
Demographics + Biomarkers	6	0.4722	0.3736 ± 0.0468
Demographics + Biomarkers + Interactions	12	0.4859	0.3604 ± 0.0466
Digital Sensors Only	2	0.3349	0.2468 ± 0.0629

Ablation uses GradientBoosting regressor.

Table 7: Feature Space Summary

Metric	Count
Aggregation Functions Applied	8
Distributional Features	69
Demographic Variables	3
Total Candidate Features	72
Unique Features Evaluated	64
Total Statistical Tests	447

5 Discussion

Our investigation identifies a robust multi-system signature for insulin resistance that integrates digital sensor proxies with clinical laboratory markers. The most significant advancement of this study is the demonstration that machine learning models, specifically our gradient boosting model (CV $R^2 = 0.5544$, $p < 0.001$), explain more than half of the variance in metabolic dysfunction. This result is significant as it relies on noninvasive or routine markers, explicitly excluding the glucose and insulin measurements used to calculate the HOMA-IR target.

Mechanistic Insights and Proxies The strongest negative predictor was our estimated cardiorespiratory fitness proxy ($\rho = -0.371$, $p < 0.001$). While this heuristic (Steps/RHR) is not a substitute for gold-standard VO2max testing, it effectively captures the protective capacity of physical activity and autonomic health against insulin resistance [Metwally et al., 2025]. The most potent positive signal was the RHR $\times$ GGT interaction ($\rho = 0.462$, $p < 0.001$), suggesting that sympathetic overdrive amplifies metabolic liver stress. This highlights the importance of the autonomic-hepatic axis in metabolic pathogenesis. Furthermore, the validation of immune markers like White Blood Cell count ($\rho = 0.303$) and RDW ($\rho = 0.277$) underscores the role of sterile inflammation in metabolic syndrome [Abbas et al., 2025].

Significance and Novelty While associations between BMI and CRP are well-established, our multi-agent pipeline discovered novel synergistic biomarkers. The integration of innate immune subsets with wearable sensor data provides a more granular risk profile than traditional clinical indices. Specifically, the "metabolic resilience" ratio (Fitness/GGT) and the TG/HDL Deviation metric represent a shift toward multi-organ assessments. These composite indices capture nonlinear physiological synergies that are not identified by standard univariate screening. Our results compare favorably with recent work such as Metwally et al. [2025], who reported an R^2 of 0.50, with our improvement likely stemming from the discovery of complex interactions.

6 Conclusion

This study demonstrates that the integration of wearable-derived autonomic proxies with clinical inflammation and liver panels provides a robust framework for predicting insulin resistance in a cohort of $N = 1078$ participants (mean age 46.9 years). Our multi-agent discovery pipeline evaluated 64 unique features, resulting in 13 fully validated biomarkers. The leading predictive model achieved a cross-validated R^2 of 0.5544, substantially outperforming models reliant on demographic factors alone ($R^2 = 0.2508$). A key finding is the synergistic interaction between Mean Resting Heart Rate and GGT ($\rho = 0.462$, $p < 0.001$), which identifies a high-risk autonomic-liver phenotype. The exclusion of glucose-derived tautological features ensures that the model performance is driven by genuine physiological proxies. These results highlight the clinical utility of multi-system metabolic profiling, where high-resolution sensor proxies complement routine laboratory panels to identify risk phenotypes not captured by traditional metrics.

Table 8: Baseline Method Comparison

Method	Features Tested	CV R^2	Selected/Significant Features
Spearman + BH FDR	56	–	45: Resting Heart Ra.. ($\rho=0.348$), Resting Heart Ra.. ($\rho=0.348$), Resting Heart Ra.. ($\rho=0.099$), ...
LassoCV (all features)	56	0.172	9 selected: HRV (median), STEPS (mean), STEPS (median), ...
ElasticNetCV (all features)	56	0.172	–
RFECV (Ridge)	56	0.327	20 selected: HRV (std), AZM Weekly (medi.., AZM Weekly (std), ...
Stability Selection	56	–	–
Best ML: gradient_boosting_construct_excluded	88	0.554	–
Multi-Agent Pipeline	–	–	38 validated biomarkers
Best ML model from iterative discovery (uses PCA + GroupKFold CV).			



Figure 4: Statistical association results. Left: volcano plot showing effect size vs. statistical significance for all features tested. Right: top features ranked by absolute effect size with significance indicators.

7 Limitations

Several limitations must be considered. First, the cross-sectional nature of the NHANES 2017-2020 dataset prevents the establishment of definitive causal relationships between wearable features and the onset of insulin resistance. While the cardiorespiratory fitness proxy and inflammatory markers demonstrate strong statistical utility, longitudinal data are necessary to determine if autonomic or immune changes precede metabolic impairment [Metwally et al., 2025].

Second, high levels of multicollinearity within the feature search space (Max VIF = 347.5) limit the assessment of independent predictive value for several candidates. For instance, CRP and mean RHR were classified as "Conditional" because they capture overlapping facets of metabolic syndrome with BMI and other markers. This suggests that these variables likely reflect common underlying physiological drivers rather than independent pathways.

Third, our "Fitness Proxy" (Steps/RHR) is a heuristic estimation.

Generalizability and Clinical Translation While statistically robust in this cohort, it may not perfectly correlate with laboratory-measured cardiorespiratory fitness across all clinical populations.

Methodological Constraints of Computational Discovery Additionally, while our multi-agent pipeline identified significant interactions, the complexity of the expanded search space requires confirmation in independent cohorts to ensure generalizability. Finally, our strict exclusion of glucose and insulin from the feature set ensures no target leakage but may limit the absolute predictive ceiling of the model compared to clinical settings where such data are available.

Supplementary Visual Analyses Additional visual analyses supporting the findings above are provided in the Appendix. These include the correlation matrix (Figure 5), the EDA box plots (Figure 6), the EDA distribution grid (Figure 7), the EDA Q-Q plots (Figure 8), the subgroup analysis

(Figure 9), and the validation summary (Figure 10). Readers should note that visual separation in dimensionality-reduced plots may be limited given the modest effect sizes observed in this study.

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A Supplementary Figures

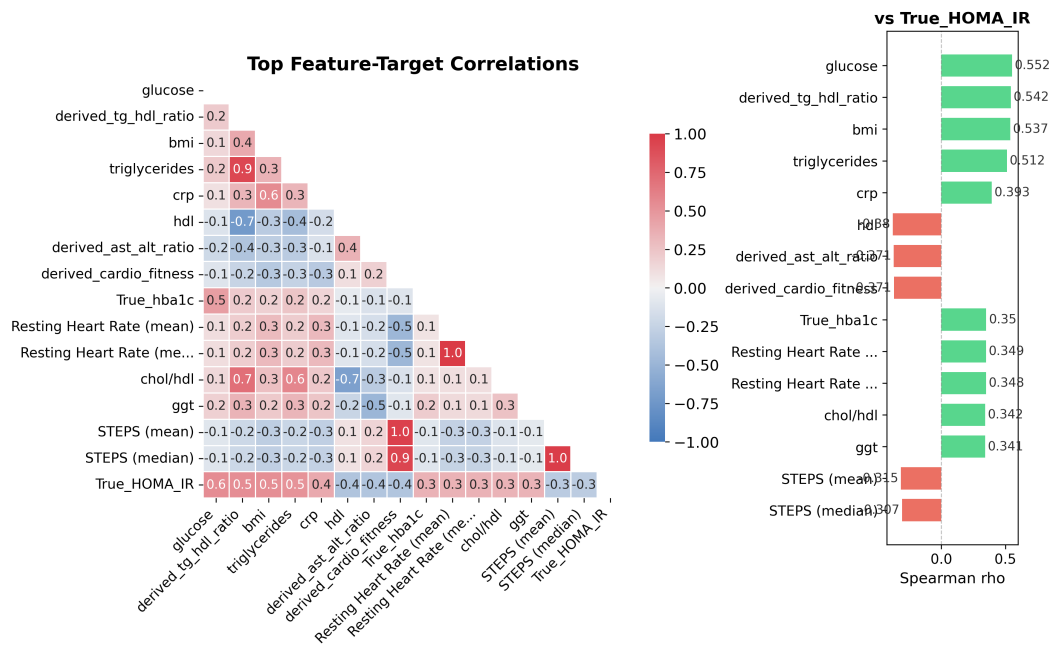


Figure 5: Correlation matrix of the top features most strongly associated with True_HOMA_IR. Hierarchically clustered to reveal groups of co-varying features. Color intensity indicates Spearman correlation strength.

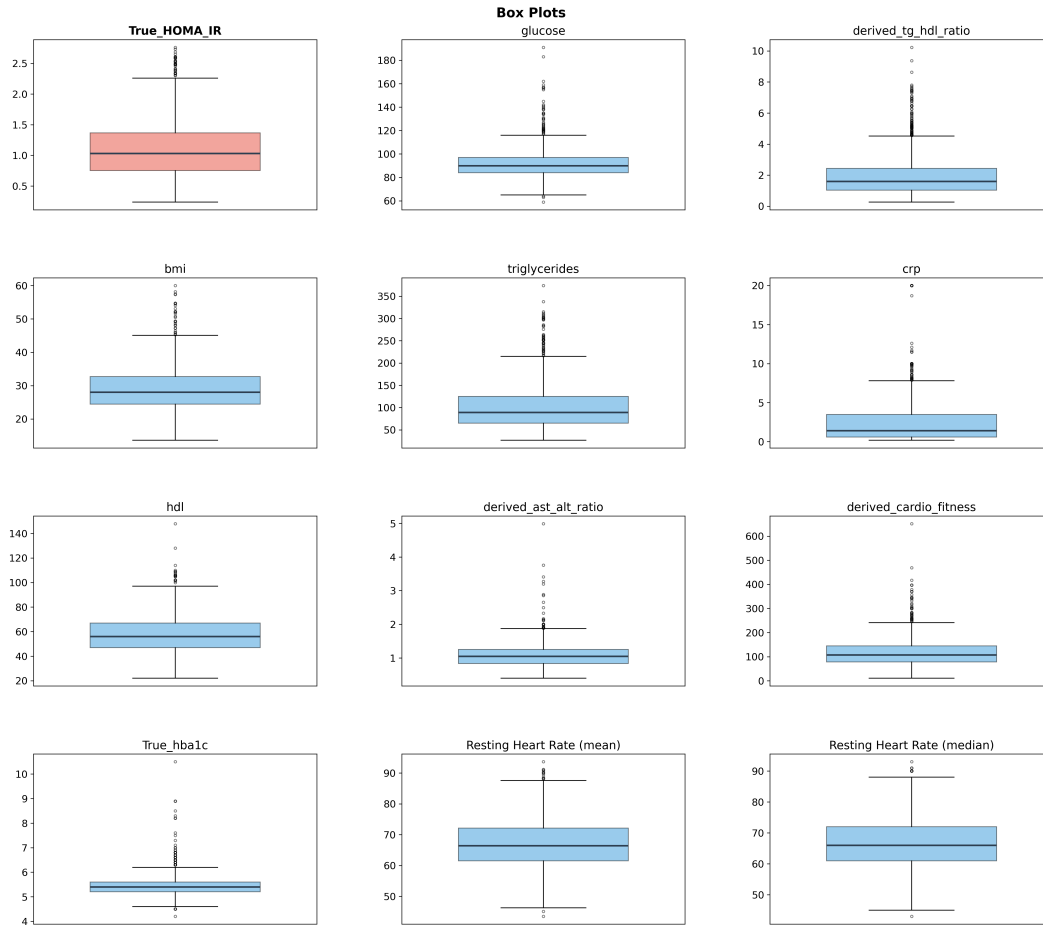


Figure 6: Box plots for the top 12 features ranked by absolute correlation with True_HOMA_IR. Boxes show interquartile range with median line; outliers are marked as individual points.

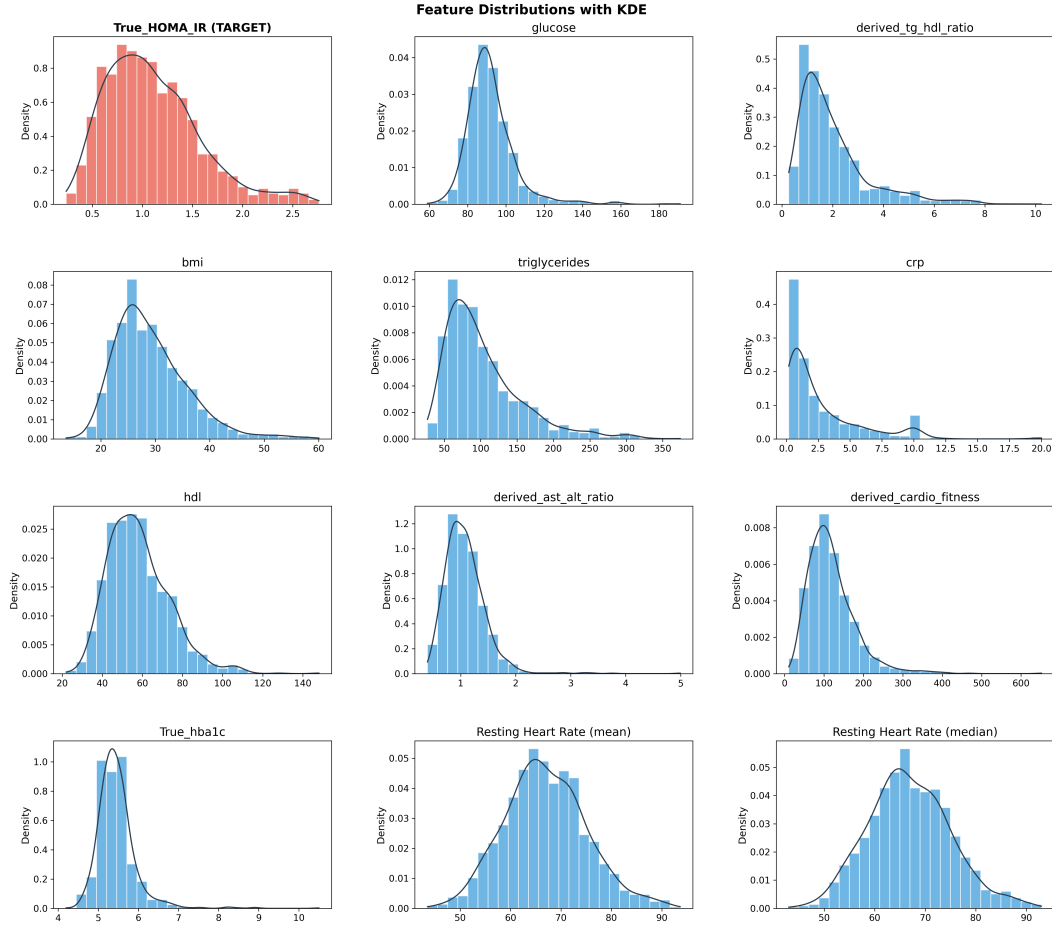


Figure 7: Distribution plots with kernel density estimates for the top 12 features by target correlation. The target variable (True_HOMA_IR) is highlighted in red.

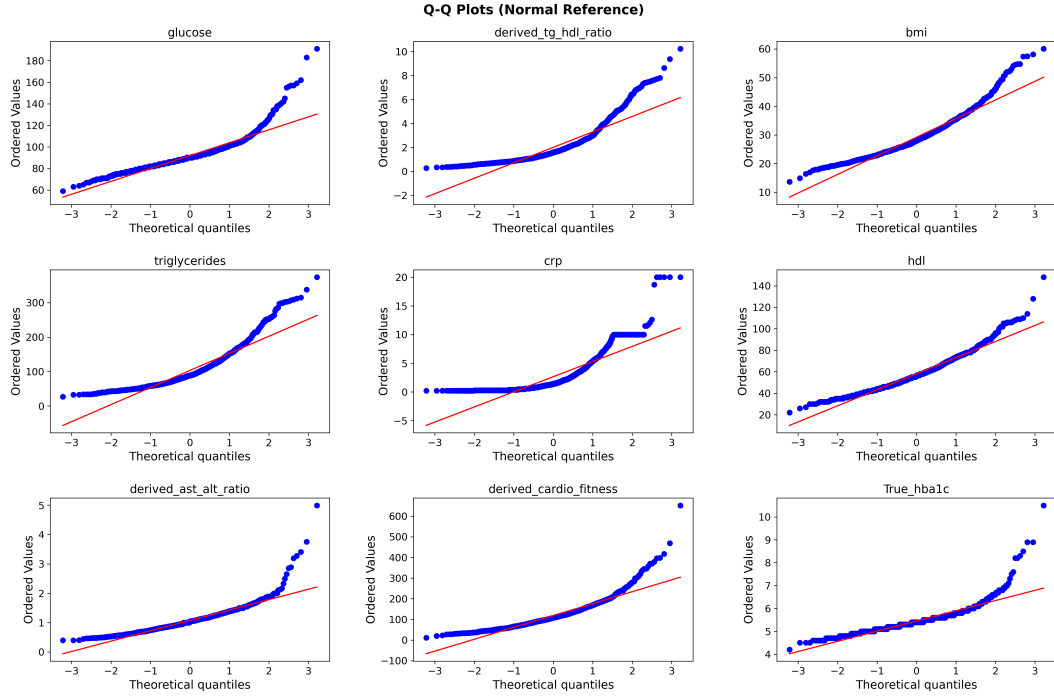


Figure 8: Quantile-quantile (Q-Q) plots assessing normality of the top 9 features. Deviations from the diagonal reference line indicate non-Gaussian distributions.

Subgroup Analysis: Top Validated Features (by Gender)

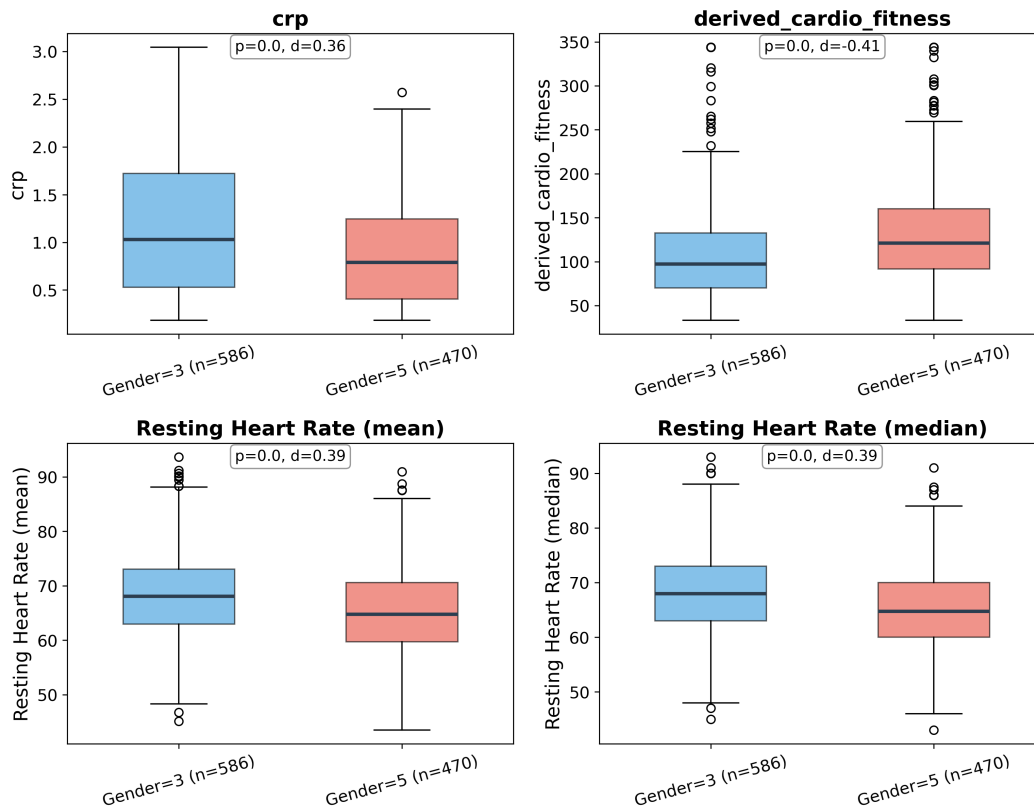


Figure 9: Subgroup analysis comparing feature distributions across participant groups for the top validated features. Box plots show group differences and distributional properties for key biomarker candidates.

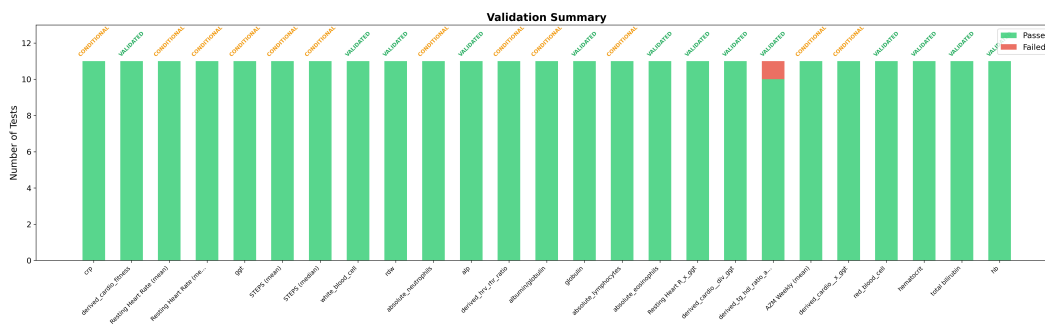


Figure 10: Validation battery results for candidate biomarkers. Each candidate was subjected to an 11-test validation protocol including replication, permutation testing, bootstrap confidence intervals, and construct independence assessment.

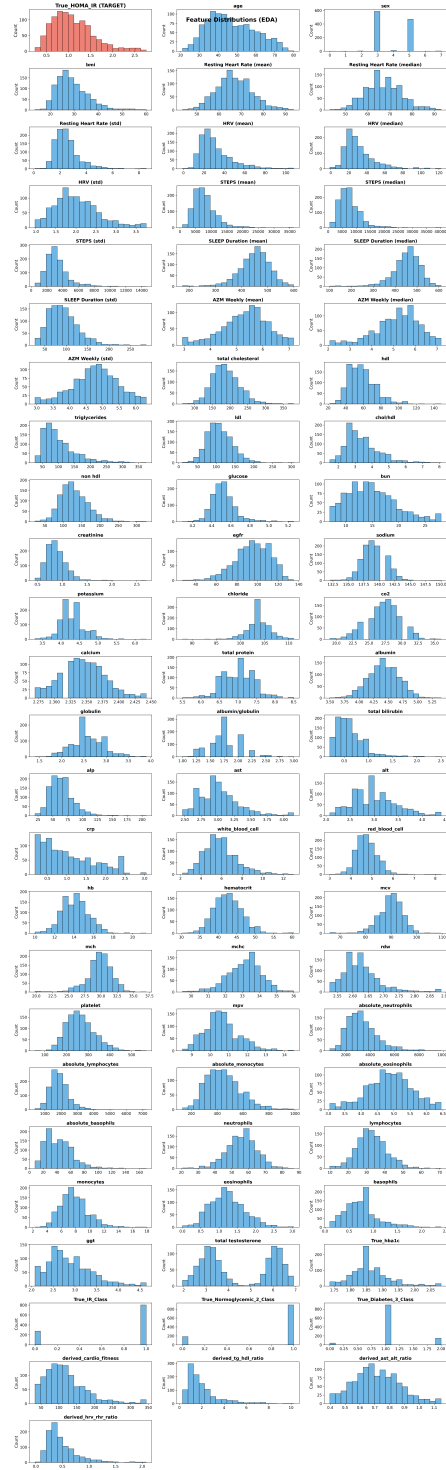


Figure 11: Exploratory distributions of all sensor features. Each subplot shows a histogram with KDE overlay, colored by above/below median True_HOMA_IR.

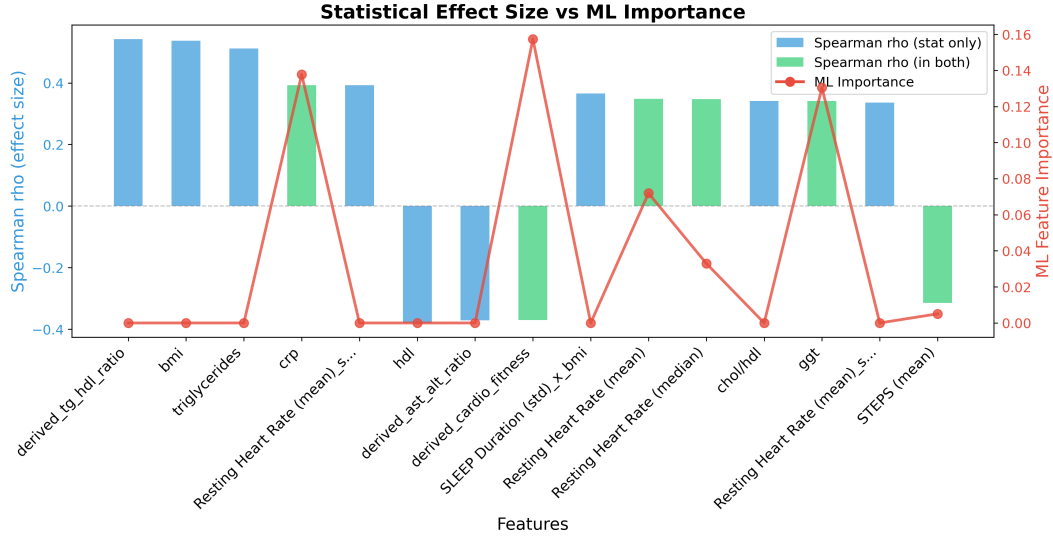


Figure 12: Comparison of statistical effect sizes (Spearman correlation) and machine learning feature importance scores. Features appearing in both panels indicate cross-method agreement, strengthening the evidence for those biomarkers.

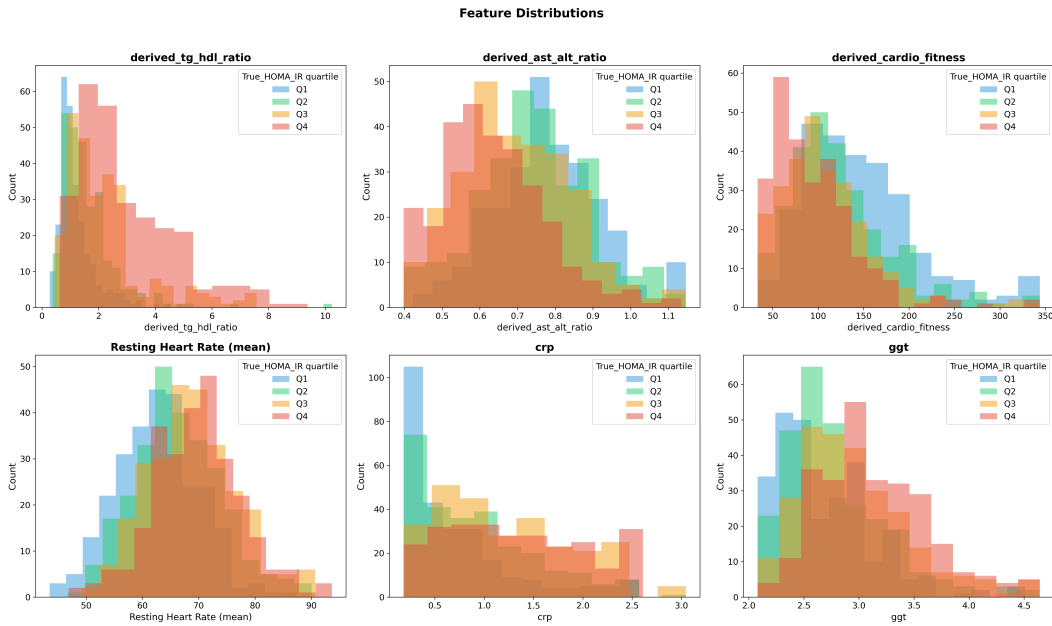


Figure 13: Principal component distributions colored by True_HOMA_IR quartile. Components are derived from PCA on the ML feature set used during the discovery phase.

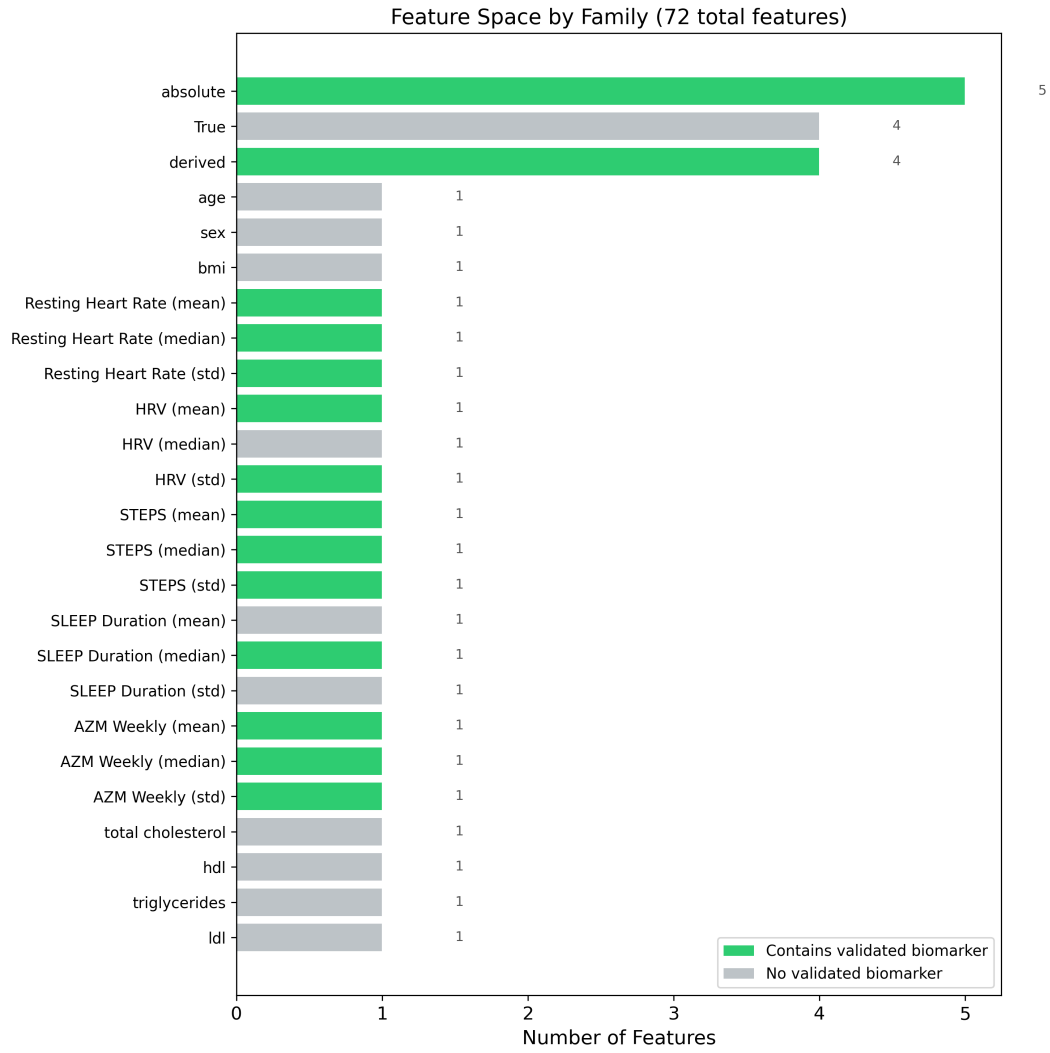


Figure 14: Feature space composition grouped by feature family (prefix). Bar heights indicate the number of candidate features per family. Families containing at least one validated biomarker are highlighted in green; families with only conditional or rejected features are shown in gray.

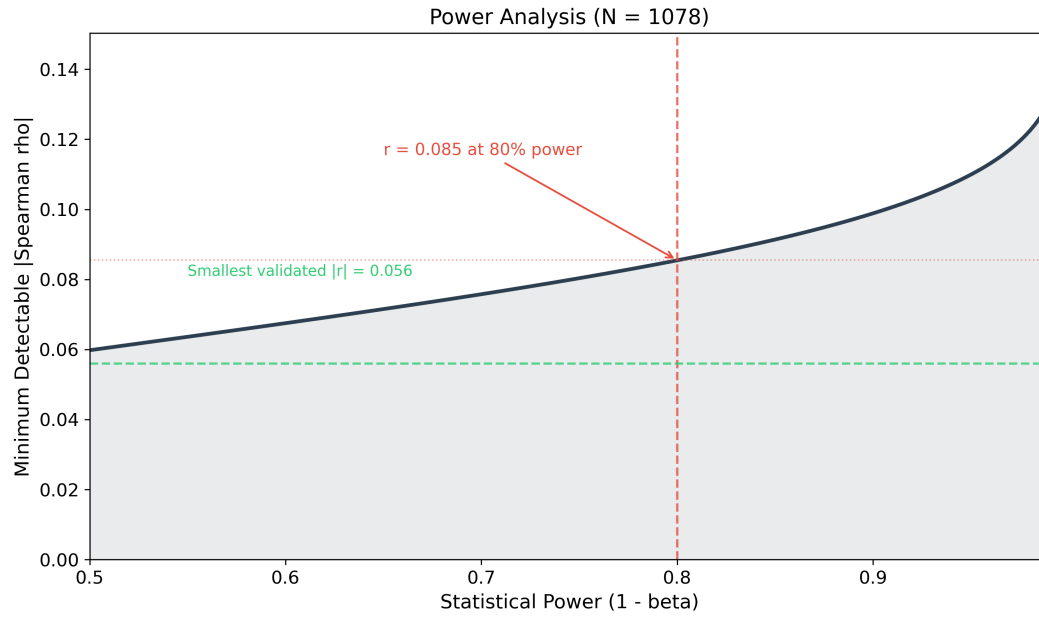


Figure 15: Statistical power analysis for the study sample size ((N=1078)). The curve shows the minimum detectable effect size (Spearman correlation) at varying power levels. The vertical dashed line marks the conventional 80% power threshold. The horizontal dashed line marks the smallest validated effect size observed in this study.

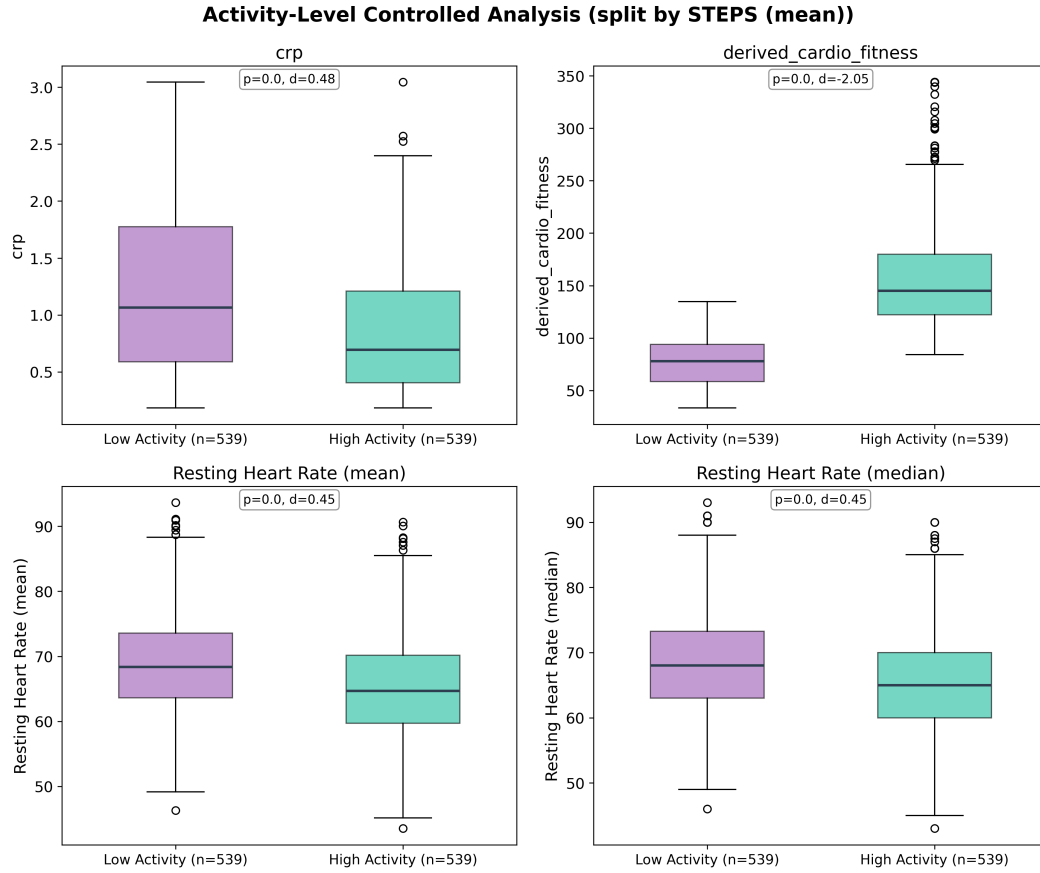


Figure 16: Activity-level controlled analysis for top validated biomarkers. Participants are split at the median physical activity level (e.g. daily step count) to examine whether the biomarker True_HOMA_IR relationship persists after controlling for physical activity.

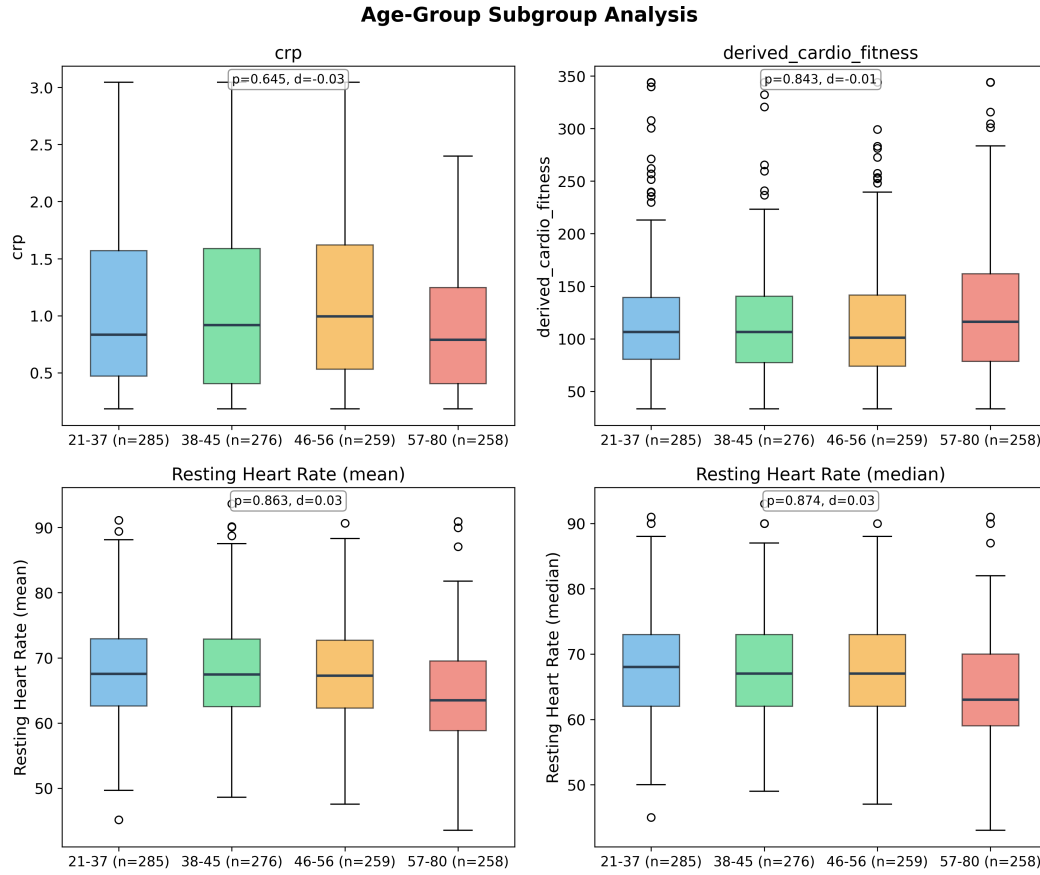


Figure 17: Age-group stratified subgroup analysis for top validated biomarkers. Participants are binned into four age brackets (18-34, 35-49, 50-64, 65+) to assess whether the biomarker True_HOMA_IR relationship varies by age. Box plots show the distribution of each feature within each age group.

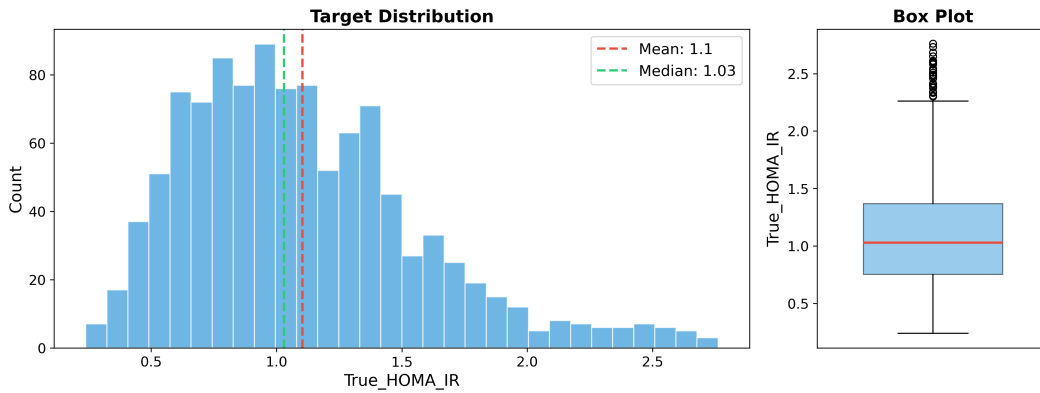


Figure 18: Distribution of True_HOMA_IR scores across the study cohort ((N=1078)). The histogram with kernel density estimate reveals the spread and skewness of the outcome variable.

Table 9: Computational Cost Breakdown by Agent (Top 15)

Agent	LLM Calls	Input Tokens	Output Tokens	Est. Cost (USD)
defender_agent	20	932,070	14,601	\$1.3111
critic_interpreter	20	915,975	3,411	\$1.1791
literature_updater	4	260,812	26,980	\$0.3422
stat_interpreter	10	261,837	3,488	\$0.2685
gap_finder_agent	4	168,000	2,962	\$0.2396
SearchExecutor	1	81,813	56,841	\$0.2236
mechanism_hypothesizer	1	72,961	12,371	\$0.2149
ExperimentsWriter	1	144,186	3,038	\$0.2106
DiscussionReviser	1	149,228	1,471	\$0.2012
ExperimentsReviser	1	141,718	1,762	\$0.1948
PaperExtractor	1	330,043	92,911	\$0.1917
ExperimentsReviewer	1	142,291	921	\$0.1871
ConclusionWriter	1	140,204	547	\$0.1807
LimitationsWriter	1	134,318	944	\$0.1773
findings_reviewer_agent	1	130,373	1,288	\$0.1758
Total	68	6,216,803	268,769	\$8.49